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Research Insights

Simplified surface-initiated iron-based photoinduced ATRP – A facile route for polymer brush synthesis in microliter volumes

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ABSTRACT

This report concerns the development of a surface-initiated iron-based photoinduced ATRP (SI-photo-Fe-ATRP) for efficient synthesis of polymer brushes under mild conditions utilizing microliter volumes of reagents. We proposed a simplified reaction setup composed of only initiator decorated substrate, monomer, solvent and $FeBr_3$ which does not require the use of any additional ligands such as amines like in copper-based SI-photoATRP or halide anions as in previous report concerning SI-photo-Fe-ATRP. This was possible due to the use of solvents that play a dual role, i.e. creating a reaction medium and complexing iron salts. SI-photo-Fe-ATRP was proven to be effective in the synthesis of thick poly(benzyl methacrylate) brushes in a reasonable timeframe under harmless visible light. As a result, the developed methodology complies with the requirements of green chemistry in many aspects e.g. by reducing complexity, minimizing waste production, using less harmful biocompatible catalyst systems and ensuring mild synthetic conditions. SI-photo-Fe-ATRP was investigated in terms of selecting optimal: monomer to catalyst ratio, surface-grafted initiator structure, light intensity, and solvent type. This work also addresses the reproducibility of polymerization and provides tips for dealing with this issue. It was shown that SI-photo-Fe-ATRP can be adopted to produce polymer brushes outdoors under sunlight and without the need to use restrictive synthetic conditions.

1. Introduction

Atom transfer radical polymerization (ATRP) is one of the most frequently studied representative of reversible deactivation radical polymerizations (RDRP). This powerful technique allows the synthesis of various polymer structures, including block copolymers, branched polymers, and polymer brushes [1]. At the time of its invention, Cu-based ATRP (historically the oldest variety of ATRP) was a huge breakthrough in the field of polymer chemistry because of its ability to tame reactive radicals, extend their apparent lifetime, and force them to propagate with monomers in a controlled manner [2]. This was achieved by reversible activation and deactivation cycles of the growing chains governed by a transition metal-based catalyst. As a result, ATRP polymerization has made it possible to obtain polymers with predefined molecular weight, low dispersity, and chain-end activity. However, conventional ATRP required high concentrations of copper salts (e.g., CuBr) to effectively maintain the equilibrium between dormant and active chains. Therefore, it quickly became apparent that polymers produced by conventional ATRP will demonstrate limited applicability in highly restrictive and sensitive to contamination biomedicine as well as electrical applications [3]. In response to that, numerous ATRP techniques aimed at minimizing the concentration of Cu-based catalysts were developed. It was achieved by regeneration of the activator by the use of e.g. radical chemical reducing agents (ICAR-ATRP) [4], chemical reductants (ARGET ATRP) [5], zero-valent metals (SARA ATRP) [6], or electrochemical reactions (eATRP) [7]. All of these methods allowed the use of low concentration of stable transition-metal compounds on higher oxidation states instead of oxygen-sensitive salts with lower oxidation state metal centers as in conventional ATRP. These notable improvements enabled the synthesis of various biocompatible materials under mild conditions, however, the need to add some extra components to the reaction solution increased the complexity of the experimental setup. Moreover, the implementation of the above-mentioned ATRP techniques to synthesis of polymer brushes by grafting from approach very often needs addition of sacrificial initiator which is required to maintain the control over the process [8].

Light-mediated ATRP techniques such as photoinduced ATRP (photoATRP) and organocatalyzed ATRP (O-ATRP or metal-free ATRP) are

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gaining increasing interest, due to their simplicity, oxygen tolerance and biocompatibility [9–11]. These techniques together with ARGET ATRP [12–14] also appear very promising for large-scale polymer brush synthesis under mild conditions, opening the door to readily available next-generation responsive coatings for high-tech applications. There are several reports presenting successful utilization of surface-initiated O-ATRP (SI-O-ATRP) in microliter volumes for production of polymer brushes based on a bunch of commercially available monomers [15,16] as well as quite complex monomers bearing iridium complexes [16], thiophene [17,18], acetylene [17] and pyridine groups (problematic in Cu-based ATRP) [19]. Surface-initiated photoATRP (SI-photoATRP) is also very powerful in terms of synthesis of different types of surface-grafted polymers using microliter volumes and a sandwich-like reaction system [20–22]. In addition to that, light-mediated ATRP methods make it possible to obtain patterned polymer films useful e.g. in biological [22] or electrical applications [17,23].

Copper-based catalysts are the most frequently investigated in terms of photoATRP [24–27] and SI-photoATRP [21,28]. Iron-based catalysts were also found to be very attractive for photoATRP due to their low cost, abundance, low toxicity, and thus biocompatibility [29–33]. Moreover, Pan et al. have proved that $FeBr_3$ may act as deactivator, as well as a source of activator ($FeBr_2$) and initiator, allowing realization of simplified photo-Fe-ATRP in solution by using only monomer, solvent and $FeBr_3$ [30]. We have recently shown that iron-based SI-photoATRP (SI-photo-Fe-ATRP) utilizing 200 ppm of $FeBr_3$ in the presence of tetrabutylammonium bromide ($FeBr_3$/TBABr catalyst) allowed controlled grafting of poly(methyl methacrylate) (PMMA) brushes from planar inorganic substrates [34]. We demonstrated that SI-photo-Fe-ATRP could be effectively controlled with or without sacrificial initiators [34]. However, to the best of our knowledge SI-photo-Fe-ATRP was not explored so far in terms of polymer brush synthesis in microliter scale.

In response to this, we present here for the first time the synthesis of model polymer brushes via SI-photo-Fe-ATRP utilizing microliter volumes of polymerization mixture. A simplified reaction setup was introduced including only monomer, solvent and $FeBr_3$, without the need of using any additional amine-based ligands as in case of copper-based SI-photoATRP [20,21,28] or halide anions as previously reported for SI-photo-Fe-ATRP in milliliter scale [34]. The conditions for the rapid synthesis of relatively thick brushes were achieved utilizing visible light, offering a complementary perspective to previous reports utilizing often higher energy UV light in SI-photoATRP. Moreover, the developed methodology considers principles of green chemistry by reducing complexity, limiting waste production, utilizing less harmful catalytic systems and ensuring mild synthesis conditions. The influence of monomer to catalyst molar ratio, surface initiator structure, light intensity and type of solvent on polymerization kinetics were investigated as these factors have not yet been studied for SI-photo-Fe-ATRP. Moreover, the main sources affecting the repeatability of the syntheses were revealed and verified. Chain-end fidelity was tested while high adaptability of the SI-photo-Fe-ATRP conditions was proved by preparation of thick poly(benzyl methacrylate) (PBzMA) brushes under nonrestrictive conditions by performing the synthesis outdoors under sunlight. PBzMA brushes, as exemplary hydrophobic coating, may be used in various applications, such as antifogging, liquid-repellent, or self-cleaning surfaces [35], drag-reduction coatings, and oil-based lubrication interfaces [36].

2. Materials and methods

2.1. Materials

Silicon wafers were purchased from ON Semiconductor (Roznov, Czech Republic). (3-Aminopropyl)triethoxysilane (APTES, 99 %), α-bromoisobutyryl bromide (BIB, 98 %), triethylamine (TEA, ≥99.5 %), toluene (HPLC Plus, ≥99.9 %), acetonitrile (MeCN, HPLC/GC, ≥99.8 %), propionitrile (PrpCN, GC, ≥99.0 %), n-butyronitrile (BuCN, GC, ≥99.0 %), dichloromethane (DCM, p.a., ≥99.9 % (GC)), methyl methacrylate (MMA, contains ≤ 30 ppm MEHQ as inhibitor, 99 %), aluminium oxide 90 active basic (activity stage I, 0.063–0.200 mm) and molecular sieves (4 Å) were purchased from Merck (Burlington, USA). Iron(III) bromide (anhydrous, ≥98 %) was purchased from Alfa Aesar (Ward Hill, USA). Benzyl methacrylate (BzMA, contains 50–100 ppm MEHQ as inhibitor, 98 %) was purchased from Thermo Fischer Scientific (Waltham, USA). α-bromophenylacetyl chloride (BPA-Cl, 95 %) was purchased from AK Scientific (Union City, USA). *N*-methyl-2-pyrrolidinone (NMP, 99.8 %) was purchased from Fluorochem (Glossop, UK). Ethanol (96 %, p.a.), methanol (p.a.), toluene (p.a.), tetrahydrofuran (THF, p.a.), hydrochloric acid (35–38 %) were purchased from Chempur (Piekary Śląskie, Poland), and dichloromethane (DCM) from Stanlab (Lublin, Poland). HPLC/GC grade solvents for initiator grafting and polymerization were dried under molecular sieves, while DCM was additionally distilled. The monomers were purified by passing them through basic alumina column to remove inhibitors. Other reagents were used as received.

2.2. Methods

Polymerization was performed under specially designed home-made photoreactor containing LED lamps with maximum emission at 450 nm (radiation intensity 94 ± 3 W m^{-2} unless stated otherwise, measured by the Delta OHM HD2302.0 light meter equipped with the probe sensitive for the spectral range 400–1050 nm). Surface topography measurements were performed using AFM Dimension ICON microscope (Bruker, Santa Barbara, USA) in PeakForce QNM® mode, using ScanAsyst-air probes (nominal spring constant 0.4 N m^{-1}). To measure the brush thickness, the brushes were scratched with a needle, washed in ultrasounds in toluene, and dried in the stream of nitrogen. AFM images were captured at the edge of the scratch. AFM measurements of rupture separations (L_R) and equilibrium thicknesses (L_E) of PBzMA brushes were performed using SNL-10 probe (Bruker, nominal spring constant 0.12 N m^{-1}, nominal tip radius 2 nm). The spring constant of the used probe was calibrated by the thermal-noise method [37]. The deflection sensitivity of the optical beam-detection system was calibrated on nondeformable silicon wafer. Force-separation curves were captured in toluene (good solvent for PBzMA). The maximum loading force (12 nN) was optimized to ensure the complete compression of the brushes reaching the constant compliance regime. Around 95 force-separation curves were analyzed to calculate the mean values of L_E and L_R. Ellipsometry measurements were performed on the spectral ellipsometer M200U (J. A. Wollam, USA). Every single sample was measured in two places and the thicknesses were determined using the Cauchy model.

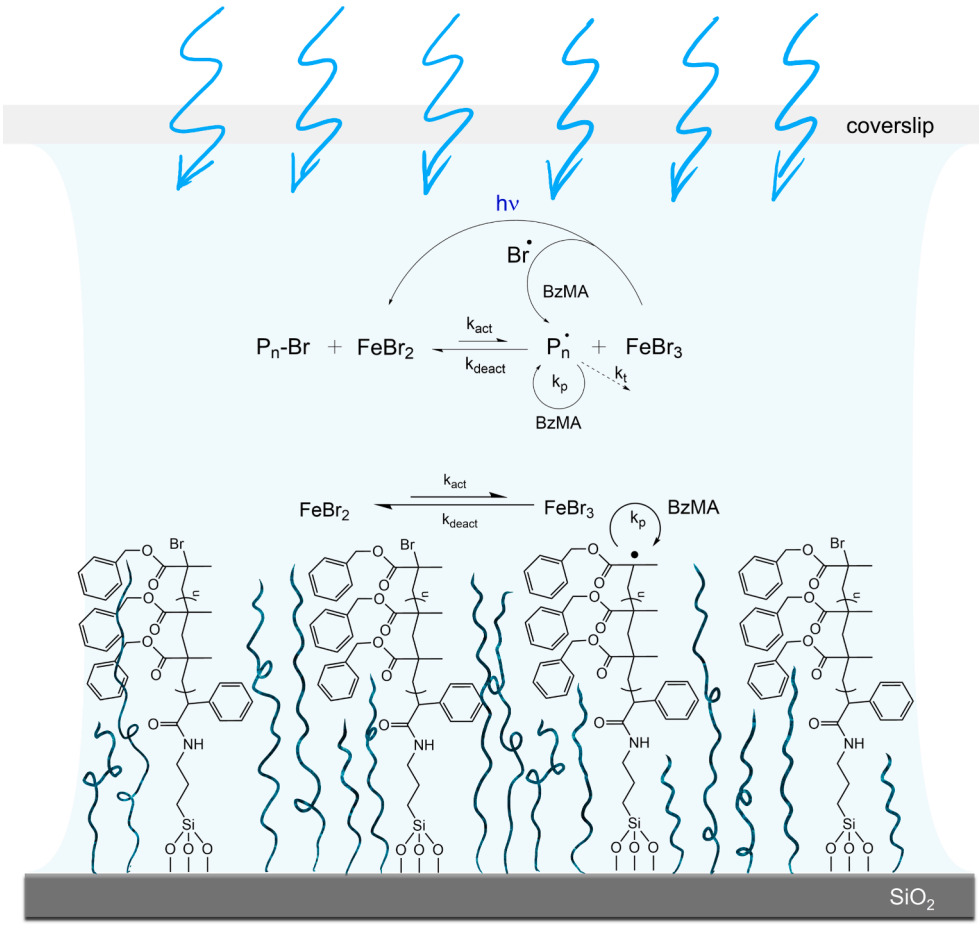
2.3. Procedures

2.3.1. ATRP initiator grafting

Silicon wafers were rinsed with ethanol, placed in the ultrasonic bath in ethanol at elevated temperature (50°C) for 10 min and dried under the stream of nitrogen. The surface was activated in the UV-Ozone Cleaner (Ossila, UK) for 30 min. The plates were placed in the conical flask containing dried toluene (10 mL), sealed with rubber septa and purged with argon. Then, APTES (10 μL) was added dropwise and the reaction was left overnight. The plates were rinsed with toluene, placed in the ultrasonic bath in toluene (10 min) and rinsed with dichloromethane, such modified substrates were then modified with either BPA-Cl forming APTES-BPA or BIB to obtain APTES-BIB initiator layer according to reported procedures [19,34,38]. In case of synthesis of APTES-BPA the reaction time was reduced to 1 h compared to previously reported conditions [34].

2.3.2. Synthesis of poly(benzyl methacrylate) brushes via SI-photo-Fe-ATRP

Stock $FeBr_3$ solution with molar concentration of 0.12 M was



Scheme 1. Schematic representation of SI-photo-Fe-ATRP of benzyl methacrylate in a sandwich-like reaction system.

prepared by dissolving $FeBr_3$ (8.72 mg, 0.03 mmol) in anhydrous MeCN (250 μL) under argon atmosphere. Polymerization solution was prepared by mixing of 225 μL MeCN, 25 μL stock $FeBr_3$ solution and 500 μL of BzMA (3 mmol), forming final polymerization solution in acetonitrile ([BzMA] = 4 M, [BzMA]: [$FeBr_3$] 100: 0.1 (mol/mol), BzMA: MeCN 2: 1 (v/v)). The reaction solution was purged by argon using the system of two amber vials sealed with rubber septa and connected via double-tipped needle. The first vial contained pure MeCN to saturate the system with solvent vapors. After argon purging, the polymerization solution was stored for 1 day in the refrigerator. Before the polymerization, such prepared solution was kept for ca. 15 min at room temperature. *It should be noted that solution storage can be probably skipped. This procedure was applied as preliminary results indicated improved polymerization kinetics. However, the tests repeated at a later stage did not clearly confirm these conclusions.* Then, 15 μL of polymerization solution was placed on the ATRP initiator-modified silicon plates (with dimensions ca. 15 x 15 mm) and covered with coverslips (18 x 18 mm). The samples were irradiated with visible light ($\lambda = 450$ nm, light intensity 94 ± 3 W m^{-2}) using homemade LED reactor for a set time. After the polymerization, coverslips were removed, and the brushes were rinsed and sonicated (5 min) in toluene and THF. Then the samples were washed with deionized water, sonicated in 0.01 M HCl, once again washed by deionized water, rinsed with methanol, and finally dried under the stream of nitrogen. To prepare polymerization solutions with different concentrations of $FeBr_3$, proportionally different volumes of stock $FeBr_3$ solution were added to the polymerization mixture while maintaining constant total volume of MeCN (250 μL). Polymerizations performed using other solvents (PrpCN, BuCN and NMP) were conducted in the same way as described for MeCN.

2.3.3. *Synthesis of poly(methyl methacrylate) brushes via SI-photo-Fe-ATRP*

Polymerization solution was prepared in the system of three amber vials connected with double-tipped needles. The first vial contained pure acetonitrile to saturate the system with solvent vapors; the second vial contained the solution of $FeBr_3$ (13.87 mg, 0.05 mmol) in acetonitrile (250 μL); and the third vial contained methyl methacrylate (500 μL, 5 mmol). After purging the system with argon, the solution of $FeBr_3$ was transferred to the vial containing the monomer, resulting in 6 M MMA, 0.06 M $FeBr_3$ polymerization solution in acetonitrile ([MMA]: [$FeBr_3$] 100: 1 (mol/mol), MMA: MeCN 2: 1 (v/v)). The polymerization solution was placed in the refrigerator for 1 day. The polymerization procedure was the same as described for benzyl methacrylate. To prepare polymerization solution with [MMA]: [$FeBr_3$] 100: 0.1 (mol/mol), stock 0.19 M solution of $FeBr_3$ (13.87 mg, 0.05 mmol) in acetonitrile (250 μL) was prepared, and further steps were the same as described for benzyl methacrylate.

3. Results and discussion

Silicon wafers were modified with ATRP initiator in a two steps procedure similarly as previously described in the literature [34,38]. In the first step APTES was grafted to the silicon wafers and then the amine groups in APTES were reacted with α-bromoisobutyryl bromide (BIB) or α-bromophenylacetyl chloride (BPA-Cl) to form APTES-BIB or APTES-BPA initiator layer, respectively. The modified substrates were then used for the grafting of PMMA or PBzMA brushes in μL scale. In a typical synthesis, 15 μL of reaction mixture, consisting of only monomer, catalyst and solvent, was placed on a silicon wafer and covered with a coverslip forming sandwich-like reaction system with polymerization

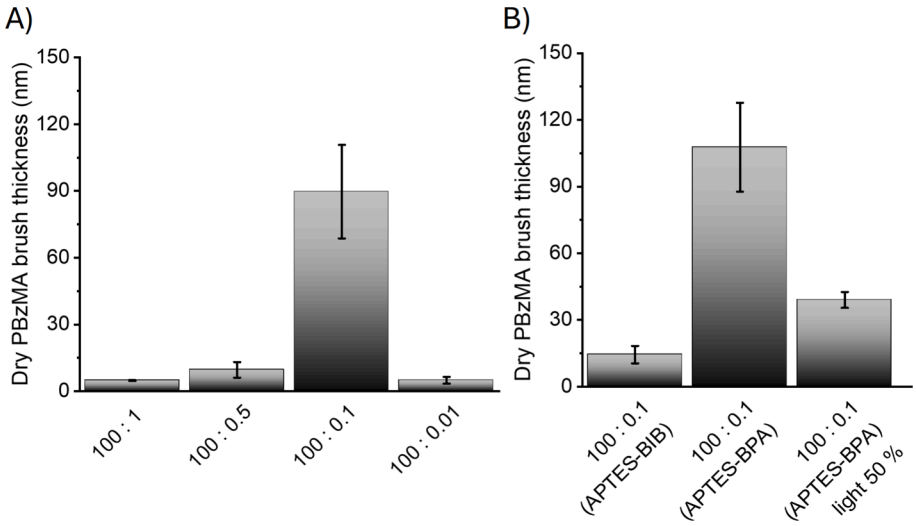


Fig. 1. A) Column graph presenting dry thicknesses of PBzMA brushes obtained after 2 h of polymerization using different monomer to $FeBr_3$ molar ratios ([BzMA]: [$FeBr_3$] = 100: x, where x = 1, 0.5, 0.1, 0.01). B) Column graph presenting dry thicknesses of PBzMA brushes obtained after 4 h of polymerization using the same monomer to catalyst ratio, while changing initiator type (APTES-BIB or APTES-BPA) and light intensity.

mixture placed in between two flat surfaces (see Scheme 1). SI-photo-Fe-ATRP was initiated by means of visible light source with emission maximum at 450 nm. Such blue light was reported to be effective for the synthesis of PMMA brushes via SI-photo-Fe-ATRP in milliliter scale utilizing $FeBr_3$/TBABr catalyst [34]. Initial polymerization conditions were the same as reported by the Matyjaszewski group for the simplified photo-Fe-ATRP in solution (MMA: MeCN = 2: 1 (v/v) and [MMA]: [$FeBr_3$] = 100: 1) [30]. Under these conditions, after 4 h of polymerization, PMMA brush with the thickness of only 5 ± 1 nm was obtained. We also tried conditions with lower concentration of $FeBr_3$ (MMA: MeCN = 2: 1 (v/v) and [MMA]: [FeBr3] = 100: 0.1) obtaining qualitatively the same results (layers thinner than 5 nm). The main problem was associated with substantial evaporation of the reaction mixture, as both MMA and MeCN are volatile liquids and demonstrate a relatively low boiling temperature. Therefore, to reduce the evaporation of the reaction mixture, we replaced MMA with a high boiling point BzMA, since the role of MeCN was not simply being a solvent but also a ligand that complexes $FeBr_3$. The polymerization conditions were further optimized considering the following parameters: monomer to catalyst ratio, type of ATRP initiator, radiation intensity, and finally the type of solvent.

3.1. Monomer to catalyst molar ratio

To find an optimal monomer to catalyst molar ratio, we performed a set of SI-photo-Fe-ATRP polymerizations of BzMA in MeCN utilizing different concentrations of $FeBr_3$ and hence applying various monomer to catalyst molar ratio (from 100: 1 to 100: 0.01, Fig. 1A). The brushes were grafted from silicon wafers decorated by APTES-BPA initiator. The obtained average thicknesses are based on at least two repeated polymerizations (Fig. 1A), while in the case of conditions for [BzMA]: [$FeBr_3$] = 100: 0.1, polymerizations were repeated 9 times (that molar ratio was further selected as optimal, so we also averaged here the values from kinetic studies, see Table S1).

The thickest brushes were obtained for the monomer to catalyst ratio 100: 0.1 (Fig. 1A). 100 ppm $FeBr_3$ ([BzMA]: [$FeBr_3$] = 100: 0.01) was found to be insufficient to effectively catalyze SI-photo-Fe-ATRP as the obtained thicknesses were very low (5 ± 1 nm) after 2 h of polymerization. Furthermore, even longer reaction times did not produce thicker brushes under these conditions. It is worth mentioning that polymerization solution has contact with both moisture and oxygen for a short time before placing the cover glass, and the possibility of oxygen diffusion during synthesis is not negligible despite the limitation of this process in a sandwich-like reaction setup. Rolland et al. have shown that in Cu-based photoATRP oxygen can be consumed through two main pathways: oxidation of CuBr (generated earlier by photoreduction of $CuBr_2$/L) and reaction with initiator radicals [39]. Similar oxygen consumption mechanism is expected for Fe-based photoATRP [40]. In case of very low concentration of $FeBr_3$, radicals generated on very active APTES-BPA initiators or growing surface-grafted chains can be easier exposed to reactions with oxygen leading to irreversible termination and hence hindered brush growth. However, the exact explanation of this phenomena needs further extensive studies. Higher concentrations of catalyst also provided rather thin layers. However, in the case of the monomer to catalyst molar ratio 100: 0.5 we observed further progression of the brush thickness with polymerization time from 9 ± 4 nm (2 h) to 25 ± 4 nm after 8 h of polymerization. In case of the highest concentration of $FeBr_3$ ([BzMA]: [$FeBr_3$] = 100: 1) the brush thickness increased only by 3–4 nm by prolonging the polymerization time up to 8 h. Layadi et al. demonstrated the existence of an optimal concentration of $FeBr_3$ (5–10 mM) in μL volume synthesis of polymer brushes by surface-initiated iron(0)-mediated ATRP, while further increasing of $FeBr_3$ concentration caused enhanced deactivation and hence slowing down of the polymerization [41]. High concentration of $FeBr_3$ was previously reported for photoATRP in solution, in which a fairly significant induction period was observed due to slow photoreduction of Fe (III) species [30]. For polymerization in μL scale, no detectable induction time was observed, as the total volume of the reaction mixture was very low. Consequently, the reaction mixture was uniformly spread over the modified substrate forming thin layer effectively exposed to the incident light in opposite to classical polymerization carried out in the flask and milliliter scale. Thinner brush layers obtained at higher catalyst concentration are rather associated with more efficient deactivation and the production of large quantities of Br• radicals during $FeBr_3$ photoreduction, which may simultaneously initiate the polymerization in the solution (Scheme 1) quickly consuming monomer molecules [30]. Pan et al. have indicated that the simplified photo-Fe-ATRP does not require additional initiators, since initiating species are generated in situ, while the targeted degree of polymerization is 2.5 times larger than the initial molar ratio of the monomer to $FeBr_3$. In other words the maximum achievable degree of polymerization DP (assuming quantitative monomer conversion) for [BzMA]: [$FeBr_3$] = 100: 1 may reach 250. Therefore, the in situ generated initiating species are expected to act as sacrificial initiators in SI-photo-Fe-ATRP. However, it is worth

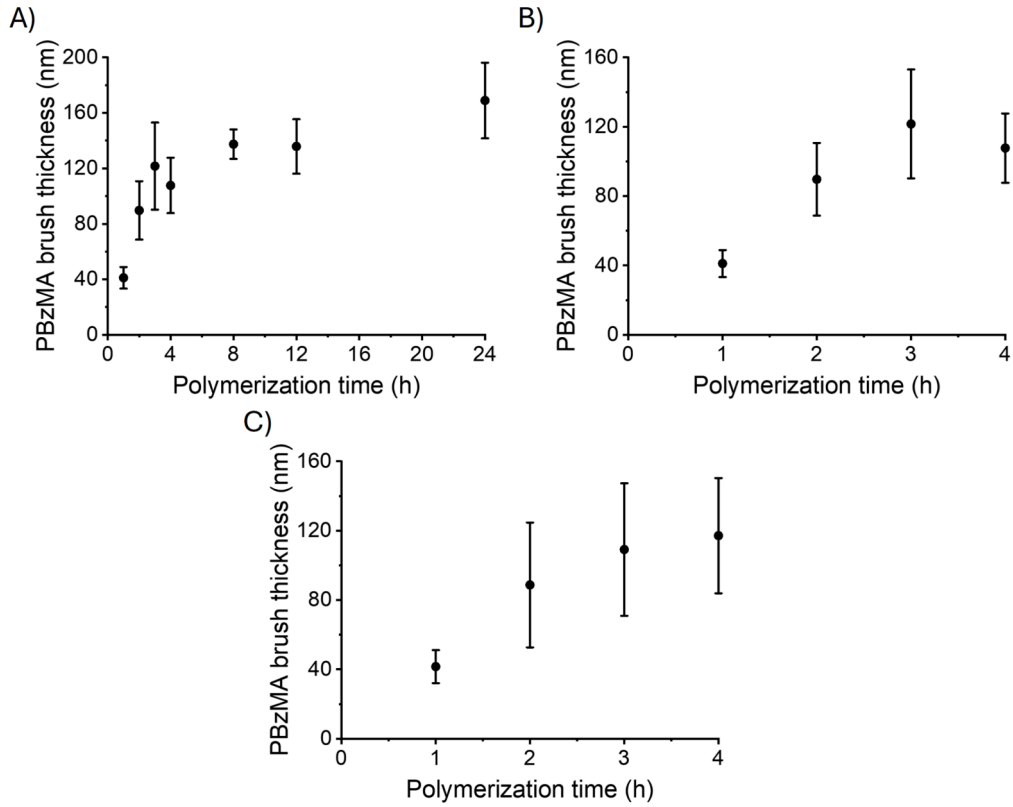


Fig. 2. Dependency of PBzMA brush thickness vs polymerization time. A) cumulative kinetic, B) cumulative kinetic up to 4 h of polymerization, C) dependency obtained only based on analysis of the samples obtained in the same series (same reaction mixture, irradiated simultaneously in the same reactor).

mentioning that the calculation of targeted DP cannot be postponed 1: 1 to polymer brushes synthesized in the current report, due to the use of surface-grafted ATRP initiators showing different activation rate constants compared to initiating species generated in the solution (structural differences). Moreover, these free initiators may initiate the polymerization with some delay compared to surface-grafted initiators because they must be in situ generated during photoreduction of $FeBr_3$ [30]. Gazzola et al. demonstrated that the addition of a sacrificial initiator to copper-based SI-photoATRP results in thinner layers when the free initiator concentration in the solution increases [21]. A comparison of SI-photo-Fe-ATRP with and without sacrificial initiator in a classical synthesis in milliliter volumes also showed the formation of thinner layers when polymerization was carried out with additional free initiator [34]. As a result, from the set of molar ratios tested, a ratio of 100: 0.1 was assumed to be optimal for the synthesis of relatively thick brushes and used for further optimization.

3.2. The influence of initiator type and light intensity

It is well known that ATRP initiators differ in terms of the activation rate constant and initiation efficiency. Yan et al. have shown that BPA-based initiators show better performance in surface-initiated metal-free ATRP compared to BIB-based ones, providing the brushes with higher grafting density and lower dispersity, due to enhanced initiation efficiency [42]. BPA-based initiators show a much higher activation rate constant due to the formation of a more stabilized radical center after the cleavage of the carbon-bromide bond. In response to that, we tested two types of ATRP initiators: APTES-BPA and APTES-BIB. Head-to-head comparison of SI-photo-Fe-ATRP initiated from APTES-BIB and APTES-BPA indicated an enhanced brush growth rate when APTES-BPA was used as ATRP initiator (Fig. 1B). PBzMA brushes grafted from APTES-BPA decorated silicon wafers were few times thicker (108 ± 20 nm) compared to the brushes obtained using APTES-BIB initiator (14 ± 4 nm) after 4 h of SI-photo-Fe-ATRP. In response to that, further kinetic studies were performed using silicon wafers decorated with the APTES-BPA initiator.

SI-photo-Fe-ATRP was found to be very sensitive to light intensity. Around 50 % reduction of light intensity (46 W m^{-2}) significantly slowed down the polymerization leading to almost 3 times thinner layers (39 ± 4 nm) compared to initial light intensity (94 ± 3 W m^{-2}, 108 ± 20 nm). Therefore, for further kinetic studies we decided to perform polymerizations using a radiation intensity of 94 ± 3 W m^{-2}.

3.3. Kinetic studies

Polymerizations were tested for up to 24 h (Fig. 2A). A nearly linear increase in dry brush thickness was observed for short reaction times of up to 3 h (see Fig. 2B). However, after a longer time the brush growth rate is notably slowing down, which is likely related to the consumption of the monomer due to simultaneous initiation of polymerization in the solution by Br• radicals forming alkyl radicals as a result of reaction with the monomer (Scheme 1). Furthermore, we observed that after 4 h of polymerization the thicknesses of the obtained brushes are markedly lower than after 3 h. All the thicknesses presented in Fig. 2A and B are based on at least 4 repeated polymerizations. However, the points assigned to 4 h were averaged based on the analysis of 15 samples. Fig. 2C presents the kinetics in which we averaged only the values gathered for the samples polymerized in the same series (same reaction mixture, irradiated simultaneously in the same reactor, see Table S1, entry 1–3). In this case, the average thickness after 4 h of SI-photo-Fe-ATRP is slightly higher compared to 3 h. However, in one of the series we obtained markedly thinner layers after 4 h. To get more insight into repeatability, we fixed the polymerization time to 4 h and ran simultaneously polymerizations for 4 samples in the same photoreactor using same reaction mixture (see Table S1, entry 4–7 and Fig. S1 presenting AFM topography images with photos of the analyzed samples). The

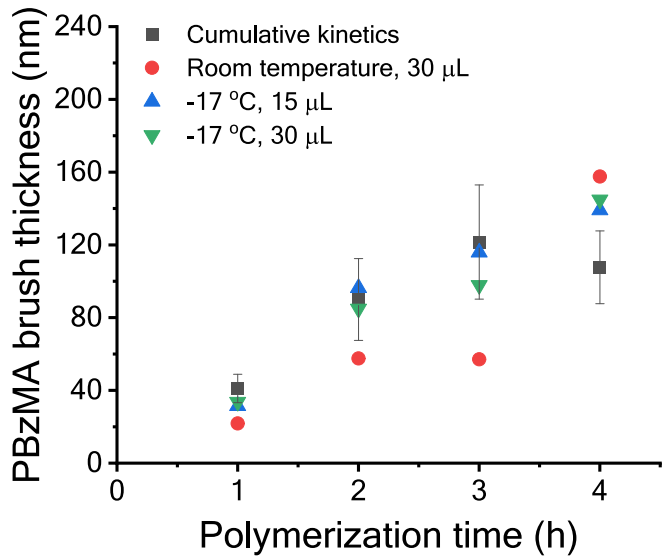


Fig. 3. The influence of volume and/or temperature of reaction mixture on dry PBzMA brush thickness vs. time of polymerization.

thicknesses obtained were in the range of 96–129 nm (mean value 115 ± 15 nm). The standard deviation was lower compared to the cumulative kinetics (Fig. 2B, 108 ± 20 nm) and the kinetics for the preselected values (Fig. 2C, 109 ± 38 nm), but repeatability was still limited. We think that this might be the effect of several factors such as 1) high sensitivity to light conditions that may slightly vary at various positions under the LED reactor, 2) uncontrolled evaporation of MeCN and exposure to oxygen and moisture especially right after putting a solution droplet on initiator modified plate, 3) positioning of the cover slip. The first and third factors were technically hard to verify. Indeed, cover glasses were not always symmetrically aligned, which might have affected the MeCN's evaporation or oxygen diffusion. However, we did not adjust the position of the coverslip after it was placed due to concerns about the possibility of spilling the solution and damaging the initiator layer.

3.4. The influence of solvent evaporation

We attempted to verify whether the evaporation of MeCN plays a substantial role. To test the influence of this factor, different attempts were made to limit the effect associated with the evaporation of acetonitrile. In the first step, we increased the injected reaction volume twice to minimize the effect of initial evaporation (Fig. 3). Unfortunately, this did not help, because the excess solution spilled outside the boundaries and some of the measured thicknesses (after 2 and 3 h of polymerization) did not well match the expected trend. In the second step, the polymerization solution was immediately injected on the initiator modified silicon wafers right after removing the mixture from the freezer (the temperature in the freezer was set at −17 °C). As a result, the initial temperature of the reaction mixture was decreased to reduce the evaporation rate of MeCN immediately after injection of the polymerization mixture on top of silicon wafer (time between injection of the solution on silicon wafer and covering it with a coverslip). Analyzing the obtained results, one may observe that cooling the initial polymerization solution can act favorably (Fig. 3), because in the case of two series of polymerizations carried out at reduced temperature, but with different volumes of solution, quite convergent results were obtained. In both cases, the increment of layer thickness after each measurement point was noticed, while the obtained dependencies were close to linear. It should be stressed that such small volumes of reagents are quickly heated to the target reaction temperature (26 °C) by the heat generated by the LED reactor. Therefore, the initial cooling down only prevents evaporation of MeCN right after injection of the reaction mixture. This result suggests that one of the main sources of the limited repeatability of SI-photo-Fe-ATRP in μL scale is probably uncontrolled evaporation of the volatile MeCN (bp 82 °C).

It is worth emphasizing that we also performed polymerizations in milliliter volumes of reagents. Polymerization was carried out for 8 h, and the layers obtained were thicker (230 ± 16 nm) compared to microliter volumes (mean thickness after 8 h 137 ± 11 nm), indicating differences between the kinetics on the microliter and milliliter scale, but also may relate to the oxygen free atmosphere in the reaction flask.

3.5. Estimation of grafting density

To test the grafting density of the PBzMA brushes, we used AFM pulling-off method to estimate the mean contour length of surface-grafted chains. PBzMA brush obtained after 2 h of polymerization with a dry thickness of 88 nm (Table S1, entry 13) was selected for this purpose. The methodology for estimating the contour length of polymer brushes by the pull-off method was described by Goodman et al.[43] It is based on the measurements of the force-separation curves in a good solvent (in this case toluene) and the analysis of the retraction curves in search of separations corresponding to rupture events (L_R). We have recently used that method for the estimation of the grafting density of the brushes produced by SI-O-ATRP [18]. When the load force of the probe is high enough (in this case 12 nN), the separation distance L = 0 nm on the force-separation curves reflects the position in which the sample behaves as an undeformable hard surface after further compression (constant compliance regime) [43,18]. In case of sharp AFM probes (the nominal tip radius according to the manufacturer's declaration is 2 nm), the separation distance L = 0 nm reflecting the

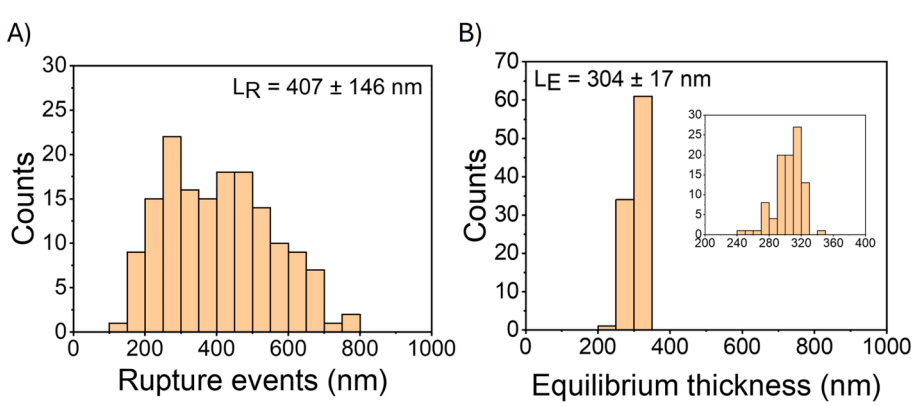


Fig. 4. Histograms presenting measured values of A) rupture separations and B) equilibrium thickness of PBzMA brushes (dry thickness 88 nm) immersed in toluene. The inset in the panel on the right side shows the distribution of the equilibrium thicknesses for smaller bin sizes for clearance.

constant compliance regime is almost equivalent to reaching the underlying hard inorganic surface by the AFM tip [43,44]. Therefore, the measured rupture separations (L_R) are approximately a measure of the lengths of the extended chain segments placed between the grafting point on silicon substrate and the rupture point at the AFM tip. Consequently, multiple rupture events in a single retraction curve (see Fig. S2) can be caused not only by the detachment of a few chains with different molar masses, but also by the detachment of a single chain at different points. However, in ca. 60 % of the analyzed force curves only one rupture separation was observed. For PBzMA brushes the mean L_R value ($\overline{L_R}$) equals 407 ± 146 nm (Fig. 4A), while the mean equilibrium thickness ($\overline{L_E}$, reflects the separation distance at which the repulsive force occurs in the approach force curve, see Fig. S2) approximately 304 ± 17 nm (see Fig. 4B). Goodman et al. demonstrated that $\overline{L_R}$ well matched the contour length (L_C) of poly(*N*,*N*-dimethylacrylamide) brushes with moderate or high grafting density in 10 mM NaCl water solution [43]. Yamamoto et al. have found that $\overline{L_R}$ of PMMA brushes with grafting density of 0.07 chains nm^{-2} immersed in toluene accounts for about 90 % of L_C of PMMA chains obtained simultaneously in the solution from sacrificial initiator [44]. However, the obtained relation was based on the assumption that surface-grafted polymers are identical with free polymers obtained in the solution. It is worth mentioning that Goodman et al. determined the L_C value of surface-grafted chains by their degrafting from latex particles and subsequent characterization by GPC excluding this source of uncertainty [43]. Therefore, we assumed that $\overline{L_R}$ is nearly the same as the L_C of surface-grafted polymers. However, we cannot exclude that the true contour length may be slightly underestimated considering the results obtained by Yamamoto et al. [44], and the assumptions that L = 0 reflects the position of the underlying surface. To calculate the grafting density Σ, the average molar mass of surface grafted chains ($\overline{M_n}$) was first calculated using formula 1, assuming that the length (l) of methacrylic-based monomer unit in fully stretched chain equals 0.26 nm as reported by Guo et al.[45]

$$\overline{M_n} = \frac{\sum_i M_i}{N} \tag{1}$$

M_{BzMA} = 176.2 g mol^{-1} – molar mass of BzMA

$$M_i = \frac{L_R}{l} M_{BzMA}$$

N – total number of analyzed chains (total number of measured rupture separations)

The calculated $\overline{M_n}$ value (275500 g mol^{-1}) and dry brush thickness (H_{PBzMA} = 88 nm) were then used for calculation of grafting density according to formula 2.

$$\Sigma = \frac{H_{PBzMA} \times d \times N_A}{\overline{M_n}} \tag{2}$$

$d = 1.18 \times 10^{-21} g\ nm^{-3}$– density of PBzMA, N_A – Avogadro's number

The achieved grafting density Σ = 0.23 chains nm^{-2} indicates that PBzMA surface-grafted chains exist in the brush regime. The obtained value is lower compared to grafting density of PMMA brushes (Σ = 0.40 chains nm^{-2}) obtained by SI-photo-Fe-ATRP in conventional way using milliliter volumes [34]. There might be multiple reasons of that, but the most important is the size of the monomer. BzMA is bulkier compared to MMA which strongly affects the packing of polymer chains and consequently grafting density as it was clearly demonstrated by Michalek et al.[46] Previously reported PMMA brushes were obtained in a sealed glass vial and oxygen free atmosphere limiting the occurrence of the side reactions with oxygen [34]. Furthermore, Σ value was derived for PMMA brushes prepared in the presence of sacrificial initiator and $FeBr_3$/TBABr catalyst [34]. Nevertheless, the obtained value is comparable to previously reported grafting densities of various brushes prepared in sandwich-like reaction systems and microliter scale: PMMA (0.28 chains nm^{-2}) [16], poly(pyridine-3-yl methacrylate) (PP3M, 0.23 chains nm^{-2}) [19], poly[2-(dimethylamino)ethyl methacrylate] (PDMAEMA, 0.1–0.2 chains nm^{-2}) [47], and poly(3-methylthiophene) (PMTM, 0.17 chains nm^{-2}) [18]. We also attempted to investigate the molecular weight distribution of surface-grafted chains according to the methodology presented by Goodman et al. by calculating weight average molecular weight ($\overline{M_w}$) (formula 3), and then dispersity (D) using formula 4 [43]. This methodology has been proven to estimate values similar to those obtained from GPC measurements. The dispersity of the analyzed PBzMA brushes reached D = 1.13 indicating formation of surface-grafted chains with narrow molecular weight distribution typical to well controlled ATRP. However, it should be noted that all rupture separations were collected locally in the center of the sample. Therefore, the dispersion covering all chains on a decorated silicon wafer is probably higher due to the oxygen diffusion causing locally hindered growth of polymer chains close to the periphery of the wafer, which is typical for microliter scale synthesis of the brushes [16].

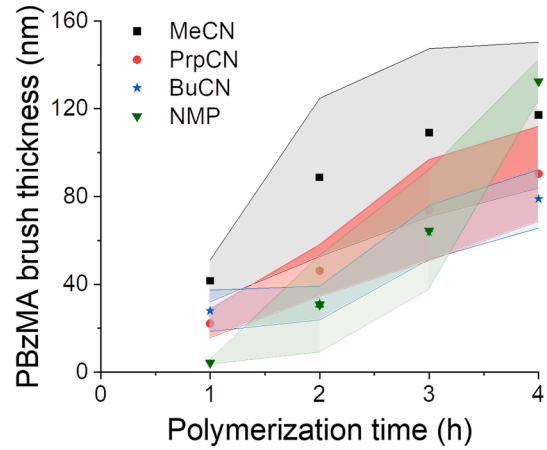


Fig. 5. Polymerization kinetics in MeCN, PrpCN, BuCN, and NMP. The graph shows the mean thicknesses captured after a certain time with error bands for clarity. Note that the results for MeCN are the same as in Fig. 2C and are shown here for a clear comparison of all solvents tested.

$$\overline{M_w} = \frac{\sum_i M_i^2}{\sum_i M_i} \tag{3}$$

$$D = \frac{\overline{M_w}}{\overline{M_n}} \tag{4}$$

3.6. The influence of solvent type

SI-photo-Fe-ATRP in μL scale was tested in solvents demonstrating higher boiling point compared to MeCN and the structure that should enable complexation of iron(III) species similarly to MeCN: *N*-methyl-2-pyrrolidone (NMP, bp 202 °C), propionitrile (PrpCN, bp 97 °C), and n-butyronitrile (BuCN, bp 118 °C). A summary of the polymerization kinetics in different solvents is shown in Fig. 5. We first tested BuCN; after running two series of polymerizations (Fig. 5) we noticed slower kinetics compared to MeCN. It might be due to the fact that bulky BuCN molecules may influence the complexing affinity of iron species and hence the polymerization kinetics. Surprisingly, when NMP was used, a kind of induction period was observed, and an acceleration of polymerization was observed with reaction time. Therefore, the resulting relationship deviated from the expected kinetics (polymerizations were repeated two times). The best compromise between the good control, repeatability and polymerization rate was achieved for polymerization in PrpCN. It seems that PrpCN can effectively form complexes with iron species and simultaneously its boiling point is higher compared to MeCN, hence

Table 1
Dry PBzMA brush thicknesses before and after chain extension experiments.

Time (h)	Dry thickness (nm)	Dry thickness after chain extension (nm)	Thickness increase
1 + 1	18	27	50 %
4 + 4	73	108	48 %

Experimental conditions: [BzMA]: [$FeBr_3$] = 100: 0.1, solvent: PrpCN.

solvent evaporation problems in uL scale could be limited. We performed 6 sets of polymerizations in this solvent and achieved a narrower distribution of the measured thicknesses compared to SI-photo-Fe-ATRP in MeCN (see Fig. 5). Therefore, we recommend using PrpCN in case of syntheses conducted in μL scale due to more predictable kinetics even though the obtained layers are thinner. In the case of classical synthesis in mL scale where repeatability problems are smaller, MeCN might be a good choice providing faster polymerization.

3.7. Chain-end fidelity and synthesis under the sunlight

Retention of chain-end functionality was tested in chain extension experiments (see Table 1). The observed ca. 50 % increase in brush thickness for PBzMA brushes obtained after 1 and 4 h of irradiation suggests the constant level of reactive end groups after the first polymerization. It additionally supports our hypothesis that after longer reaction time the brush growth rate is decreasing mainly due to substantial monomer conversion while not irreversible termination as a main reason.

To show the versatility and adaptability of the optimized polymerization conditions in PrpCN, SI-photo-Fe-ATRP was run outdoors under the sunlight for 4 h. After this time, 95 nm thick PBzMA brushes were obtained (as shown in Fig. 6). This shows that effective brush synthesis can be achieved under nonrestrictive conditions, which is necessary in terms of potential scalability of this synthesis.

4. Conclusions

The feasibility of using SI-photo-Fe-ATRP in microliter scale synthesis of model polymer brushes has been positively verified. It was shown that polymerization can be effectively realized in sandwich-like reaction systems using only monomer, solvent, and $FeBr_3$, without the need of using any additional ligands as in case of copper-based SI-photoATRP and previous reports concerning SI-photo-Fe-ATRP. However, the obtained results clearly indicated the need to modify the reaction conditions originally developed for photo-Fe-ATRP in solution. It turned out that the μL scale polymerization of a volatile monomer (MMA) in a volatile solvent (MeCN) is difficult to realize due to the rapid evaporation of the reactants and changes in the composition of the reaction mixture. In case of the reaction mixture composed of non-volatile monomer and MeCN relatively thick PBzMA brushes (100–150 nm) were easily achieved within 3–4 h of polymerization. The most efficient polymerization and fast brush growth rate were observed for monomer to catalyst molar ratio 100: 0.1. Higher concentrations of $FeBr_3$ provided thinner layers due to enhanced deactivation and high concentration of radicals produced during $FeBr_3$ photoreduction (reaction between Br• and monomer), which can simultaneously initiate polymerization in solution above the surface-grafted chains. It results in significant consumption of the monomer limiting formation of polymer chains with high degree of polymerization. On the other hand, very low catalyst concentration (100: 0.01) was ineffective and provided thin brushes. SI-photo-Fe-ATRP demonstrated limited repeatability because of the high sensitivity to light conditions and uncontrolled evaporation of MeCN, especially right after putting a solution droplet on the initiator-modified plate. The replacement of the volatile MeCN with the less volatile PrpCN increased reproducibility and allowed satisfactory polymerization rates to be maintained. Optimized conditions of SI-photo-Fe-ATRP provided PBzMA brushes with high chain-end fidelity. The use of an energy-saving LED visible light source, biocompatible catalyst, simplified reaction setup and minimization of reagent consumption are crucial for improving the eco-friendliness and reducing the complexity and costs of synthesis of polymer brushes. Moreover, it was proved that polymerization can be effectively initiated under sunlight allowing formation of thick uniform layers outdoors under non-restrictive conditions. The employed sandwich-like reaction system in SI-photo-Fe-ATRP should make it possible in the future to easily adjust polymerization conditions (using larger initiator-decorated substrates and increasing the volume of the reaction mixture accordingly) to produce stable functional coatings on large area flat surfaces.

CRediT authorship contribution statement

Anna Kiełbasa: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Data curation. **Karol Wolski:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

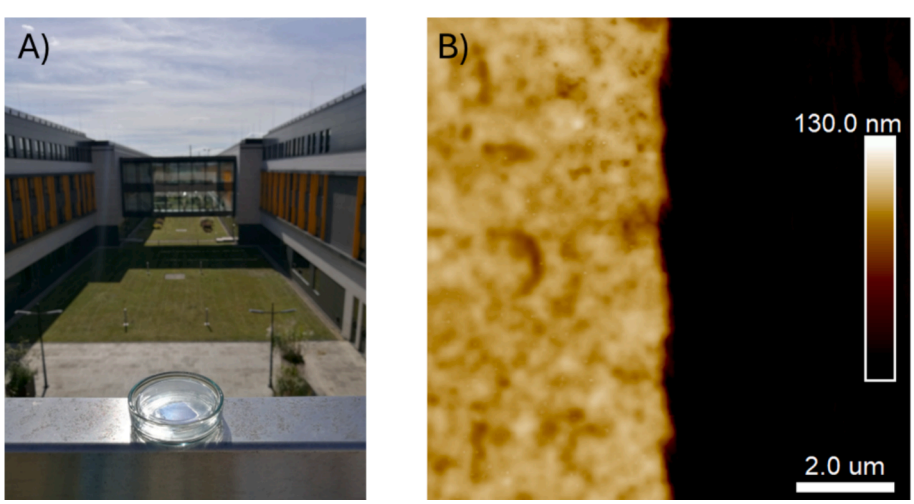


Fig. 6. A) Experimental set used in simplified SI-photo-Fe-ATRP induced by the sunlight and B) AFM topography image of the scratched PBzMA brushes (dry thickness: 95 nm). Experimental conditions: [BzMA]: [$FeBr_3$] = 100: 0.1, solvent: PrpCN, time: 4 h. Polymerization was performed in a typical September day in Poland – sunny but cloudy.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.113961.

Data availability

The data that supports the findings of this study are available in RODBUK – Cracow Open Research Data Repository (https://doi.org/10.57903/UJ/B8UGMZ).

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